

RV College of Engineering
Experiential Learning Report
Project-Based Learning

2024-25



Title of the Project

Smart Energy Monitoring And Control System

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1. Introduction

A **Smart Energy Monitoring and Control System** is an advanced solution designed to help users keep track of their electricity consumption and manage it more efficiently. With rising energy costs and growing concerns about environmental impact, monitoring and controlling energy use has become more important than ever.

This system allows users to measure real-time data such as voltage, current, power, and total energy consumed. By providing detailed insights, it helps identify unnecessary power usage and encourages more conscious consumption habits.

In addition to monitoring, the system also offers automated control features, such as turning off appliances when they are not in use, disconnecting power if an overload is detected, or cutting off supply when certain preset limits are reached. It can also include a prepaid model where users must maintain a balance to continue using electricity, further promoting responsible usage.

2. Problem Definition

2.1. Problem Statement:

In many residential, commercial, and rental setups, monitoring and controlling electricity consumption is a growing concern. Traditional energy meters do not offer real-time consumption data, nor do they provide mechanisms to limit or control energy use based on a pre-set budget. This often results in wastage, overconsumption, and billing disputes.

The need for a **smart, prepaid energy monitoring system** is essential to empower users to manage their electricity usage efficiently. By integrating a prepaid feature, users can control costs, reduce wastage, and avoid unexpected electricity bills. This system holds relevance in hostels, rental accommodations, laboratories, and small businesses where energy consumption needs close monitoring and control.

2.2. Background Information:

With the growing demand for energy conservation and efficient management, smart energy metering systems have gained considerable attention in recent years. Traditional meters only measure total consumption without offering real-time data or control over energy usage. To address this, researchers have proposed prepaid energy meters, which help consumers monitor and manage electricity within a set budget, reducing wastage and promoting cost savings.

Studies by Zain et al. (2018) and Kumar & Singh (2019) highlight the effectiveness of prepaid systems in increasing consumer awareness and reducing consumption. The integration of Internet of Things (IoT) technologies, particularly using modules like the **PZEM 004T** for accurate electrical measurements, has

further enhanced these systems. The **ESP32** microcontroller is widely adopted in such projects for its low cost, Wi-Fi connectivity, and versatility in IoT applications.

However, existing solutions often lack features like automatic power cutoff and user-friendly remote dashboards. This project aims to fill this gap by combining prepaid logic, real-time monitoring, and remote accessibility to create an affordable, smart prepaid energy management system.

3. Objectives

a. Primary Objectives:

2. To design and implement a **smart prepaid energy meter** using ESP32 and PZEM004T.
3. To enable **real-time monitoring** of electrical parameters such as voltage, current, power, and total energy consumed.
4. To integrate a **prepaid system** where users can recharge credits and automatically disconnect the load when the credit reaches zero.
5. To provide a **web-based dashboard** for users to view consumption data and manage credits remotely.
- 6.

a. Secondary Objectives:

7. To implement **data logging** for analyzing energy consumption patterns over time.
8. To introduce an **alert system** that notifies users when credit is low or in case of abnormal consumption.

3. Methodology

Approach: The project follows an IoT-based system design approach, combining hardware and software to create a real-time prepaid energy monitoring solution. The central idea is to measure energy consumption using the PZEM 004T sensor, process the data via an ESP32 microcontroller, and provide users with a web-based dashboard for credit management and monitoring. A flow chart representing the process includes key stages: power measurement, credit check, relay control, and web interface update.

The system is based on the theoretical framework of **event-driven programming** and **real-time data monitoring**, allowing automatic actions (such as load disconnection) based on predefined credit limits.

4.2 Procedures:

- **Component Selection:** Identification and procurement of components such as ESP32, PZEM004T, relay module, power supply, and necessary sensors.
- **Circuit Design:** Designing the hardware setup including sensor connections, relay wiring, and ESP32 programming.
- **Software Development:** Writing and testing Arduino code for data reading, prepaid logic, and web server implementation.
- **Dashboard Creation:** Developing the web interface to display real-time data and allow credit recharges.
- **Integration:** Combining hardware and software components into a working prototype.
- **Testing and Validation:** Performing tests to ensure accuracy of measurements, relay control, and system reliability.
- **Timeline:** The project was completed over a **4-week period**, with two weeks for hardware/software development and two weeks for testing and refinement.

5. Project Execution

5.1 Planning and Design:

The project began with a series of brainstorming sessions to define the problem, set objectives, and explore potential technologies. The team evaluated different microcontrollers and sensors, ultimately selecting the ESP32 for its IoT capabilities and the PZEM 004T for precise energy measurements. Initial design drafts were created for the hardware circuit and user interface, followed by risk assessment and safety considerations for working with AC power..

5.2 Implementation:

- Assembling the hardware components on a **breadboard** for stability.

- Developing and uploading the **Arduino code** to the ESP32 to read energy data and control the relay based on credit availability.
- Creating a **responsive web dashboard** for real-time monitoring, credit top-up, and alert notifications.
- Integrating the alert system and relay control to ensure automatic disconnection when the prepaid credit runs out.
- Conducting multiple test cycles to fine-tune the accuracy, response time, and system reliability.

6. Tools and Techniques Used

6.1 Tools:

Here are some of the list of tools we have used:-

Tools	Description
ESP32 Microcontroller	Central control unit for processing data, managing credit, and hosting the web server.
PZEM004T Module	Measures voltage, current, power, and total energy consumption accurately.
Relay Module	Controls connection/disconnection of the load based on available credit.
DC Voltage Amplifier	Amplifies control voltage from the ESP32 to reliably switch the relay.
TFT/LED Display	Displays real-time power consumption, credit, and alerts locally.
Web Dashboard (Software)	Provides users with an online interface to monitor consumption and recharge credits.
Arduino IDE	Software used for coding, compiling, and uploading programs to the ESP32.
Breadboard and Connecting Wires	Used for safe prototyping of the circuit before final assembly.

Power Supply (AC & DC)	Provides operating voltages to ESP32, sensors, and relay.
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6.2 Techniques: The following techniques and methods were applied during the development of the Smart Energy Monitoring and Control System:

1. **Embedded System Design:**

The project utilized the ESP32 microcontroller programmed using **Arduino IDE** to perform real-time monitoring, data processing, and decision-making based on energy consumption and available credit.

2. **Real-Time Data Acquisition:**

The **PZEM 004T** sensor module was employed to continuously measure electrical parameters such as voltage, current, power, and energy. This data was processed in real-time to ensure timely system response.

3. **Signal Amplification Technique:**

Since the ESP32 GPIO pins operate at 3.3V logic levels, a **DC voltage amplifier** was used to boost the signal and drive the **relay module**, ensuring reliable switching of the connected load.

4. **Event-Driven Control Logic:**

The prepaid system was implemented using **event-driven programming**, where energy consumption events trigger actions such as cutting off the power supply when credit is exhausted.

5. **Web-Based Monitoring:**

A simple **web server** hosted on the ESP32 was used to create a **remote dashboard**. This allowed users to view real-time data and recharge credits through any smart device connected to the same network.

6. **User Alert System:**

To enhance usability, an **alert mechanism** was integrated to notify users when the remaining credit falls below a predefined threshold or when abnormal consumption is detected.

7. Results and Discussion

The Smart Prepaid Energy Monitoring and Control System was successfully designed, developed, and tested. The system accurately measured electrical parameters such as voltage, current, power, and total energy consumption using the PZEM 004T module. The ESP32 microcontroller processed this data and controlled the relay to automatically disconnect the load when the prepaid credit was exhausted. A web-based dashboard provided real-time monitoring, credit management, and low-credit alerts.

The system met all the primary and secondary objectives, proving to be accurate, responsive, and user-friendly. It successfully empowered users to control energy consumption and avoid overuse, supporting cost savings and energy efficiency. The inclusion of the DC voltage amplifier ensured reliable relay switching, addressing initial instability issues. Overall, the project demonstrated strong potential for real-world applications in homes, rentals, and small businesses.

8. Prototype (Hardware/Software)

8.1 Prototype Description:

The developed prototype consists of both hardware and software components integrated into a working system. The main hardware includes:

- ESP32 microcontroller for processing and control.
- PZEM004T module for real-time power monitoring.
- Relay module with DC voltage amplifier for load switching.
- Web-based dashboard for remote monitoring and control.

Key features:

- Real-time display of voltage, current, power, and total consumption.
- Prepaid system with load disconnection when credit runs out.
- Low-credit alerts.
- Remote access via browser.

The prototype operates on both **AC and DC supply** and is designed to be compact, scalable, and cost-effective.

8.2 Development Process:

The development process began with hardware selection, followed by circuit design and simulation. After assembling the components on a breadboard, initial testing was performed to verify correct sensor readings and relay control. The software was developed in Arduino IDE, including the web interface using ESPAsyncWebServer.

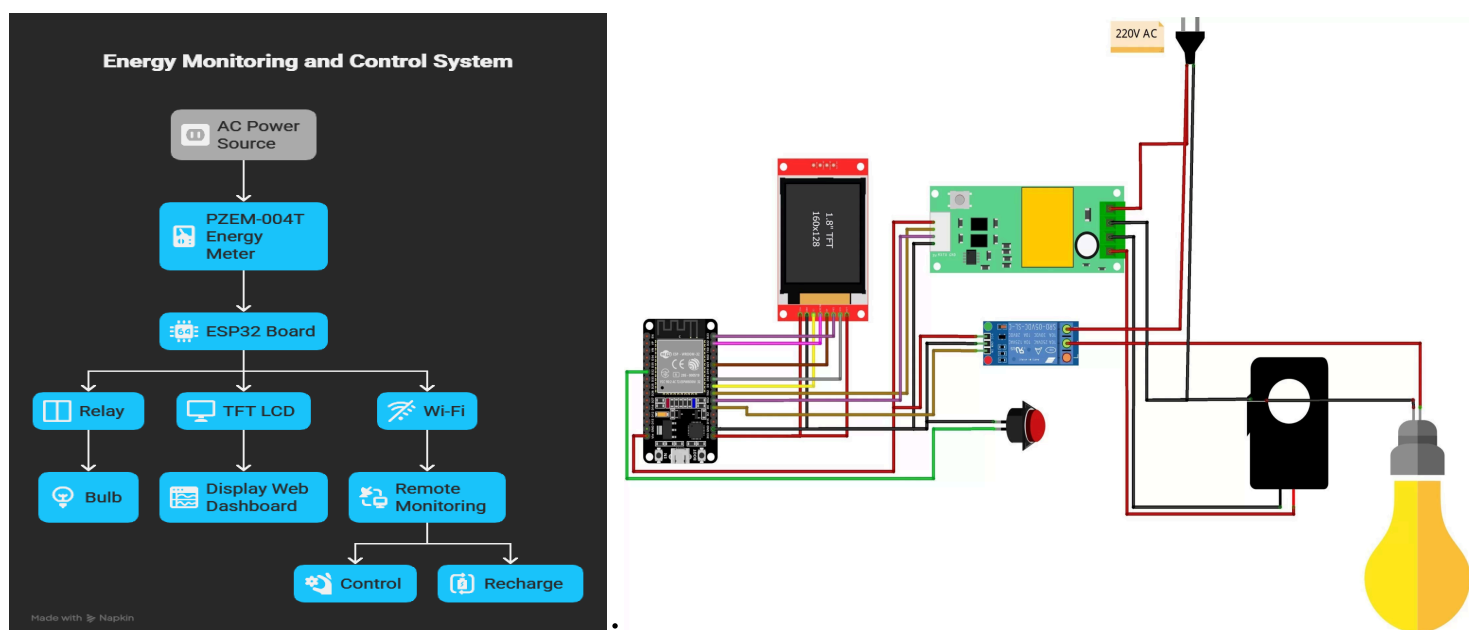
Challenges faced:

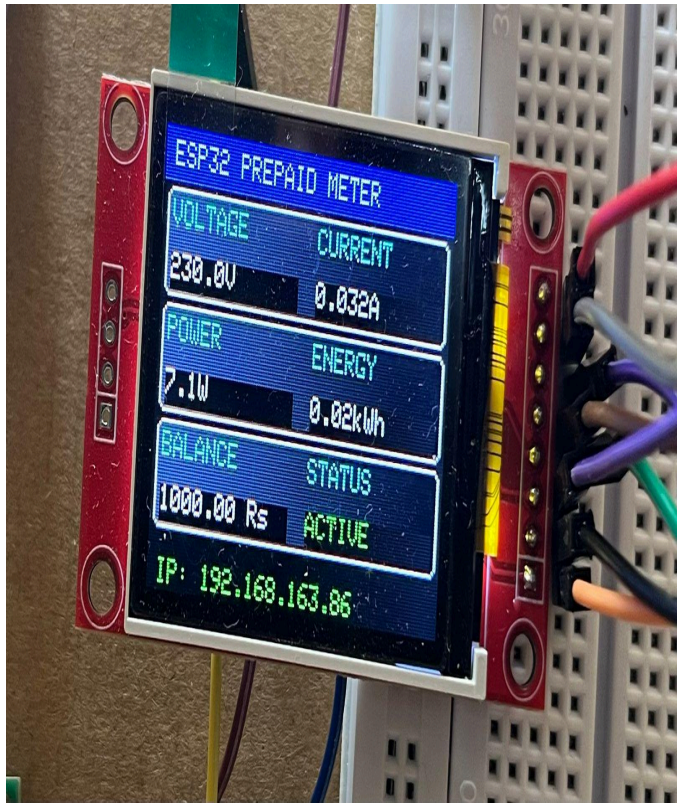
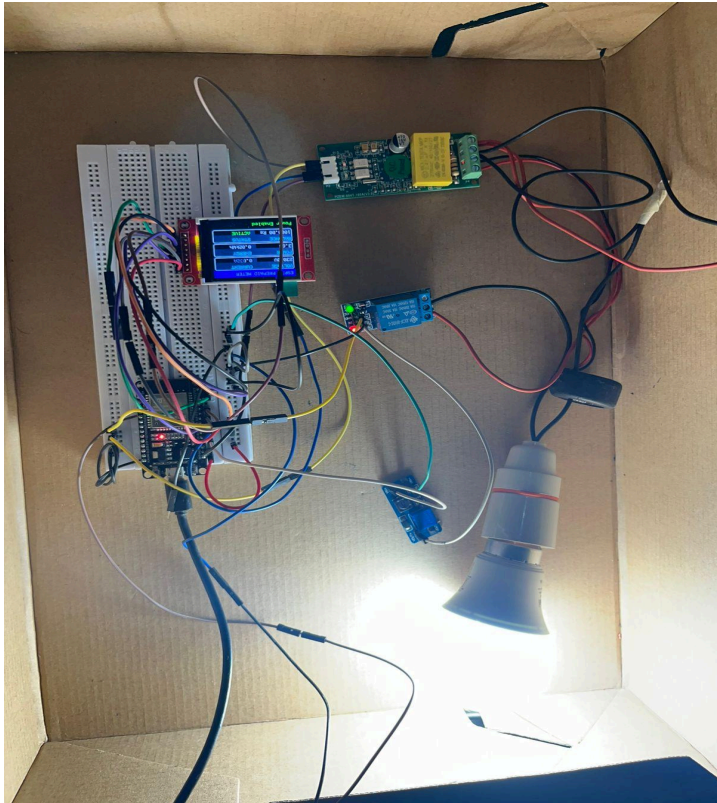
- Relay triggering instability was overcome by adding a DC voltage amplifier.
- Ensuring accurate real-time readings required careful calibration of the PZEM 004T module.
- Creating a user-friendly web dashboard with responsive design..

9. Conclusion

- In this project, we successfully designed and implemented a Smart Energy Meter with prepaid recharge functionality using the ESP32 microcontroller. By integrating components such as the PZEM-004T sensor, relay module, and an IoT-based web interface, we were able to monitor and control energy usage in real time.
- Through this project, we gained practical experience in embedded system design, IoT integration, and web-based control. We were able to automate energy management and provide an efficient way to monitor consumption and balance, enhancing user convenience and awareness.

10. Visuals





- Dashboard
- Account
- Control
- Settings
- Reports
- Help

Energy Dashboard

Real-time monitoring of your prepaid energy system

Last updated: 10:04:07 AM

BALANCE

1000.00₹

Available credit

VOLTAGE

0.0V

Line voltage

CURRENT

0.000A

Line current

POWER

0.0W

Active power

SYSTEM STATUS

Power	ON
Fault Status	OK
Theft Detection	OK
Total Energy	0.000 kWh

QUICK ACTIONS

Control power supply to your premises

Power ON

Power OFF

Quick recharge your account

₹

Recharge Now

EnergyPro

Dashboard

Account

Control

Settings

Reports

Help

System Settings

Configure thresholds and system parameters

Safety Thresholds

Over Voltage Threshold

260

V

Over Current Threshold

0.06

A

Security Settings

Theft Detection Threshold

0.020

A

Billing Settings

Minimum Balance Threshold

600

₹

Cost per kWh

0.20

₹

Save All Settings

Factory Reset

Reset all settings to default and clear energy data. This action is irreversible.

Factory Reset

Update Failed

Prepaid Energy Monitor Pro © 2025

11. QR Code of Demonstration Video

